

Lillian Hunt

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Education

University of Illinois at Urbana-Champaign

Expected May 2028

B.S. Aerospace Engineering — Minor in Computer Science

GPA: 3.63/4.00

Relevant Coursework: Computer Aided Design, Nanosatellite Design, Thermodynamics, Dynamics

Work Experience

Engineering Intern - Rolls-Royce LibertyWorks (CUI)

May 2026 – Aug 2026

- Supported legacy engine iteration project by implementing preliminary research and trade studies into additional sensor equipment to find mass flow rates through MOTS plumbing system.
- published peer-reviewed design for Venturi-style flow meter probe verified against aerodynamic performance requirements with functional design data for project
- identified 27 design iterations for tee-duct plumbing that would interfere with the pipe's natural frequencies across expected operating conditions to reduce noise within subsystem

Undergraduate Researcher - Laboratory for Advanced Space Systems

Aug 2025 – May 2026

- Recognized inadequacy of the team's 1-node thermal approximation for mission-critical decisions and led transition to a 13-node transient thermal model (Thermal Desktop/SINDA) — defining bounding hot/cold orbital cases at 400 km altitude, power dissipation models tied to operating modes, and radiative coupling/articulated conductors to characterize orientation and eclipse behavior.
- delivered a credible thermal baseline revealing avionics and solar arrays operating near survival limits — directly motivating adoption of active thermal controls and laying groundwork for the final 100+ node model in MonARCH's design.

Project Experience

Micro-g NExT (NASA Challenge Team)

Project Manager

May 2026 – Present

- Restructured leadership team and selection timelines to promote productive workflow through organization by prioritizing general member onboarding at the start of the year and mentorship.
- Organized project effort with 50 engineers split between design, verification and outreach teams to submit a proposal for a Lunar Adjustable Multipurpose Stand (LAMS) to answer NASA's call for proposal.

Design and Manufacturing Lead

Sep 2025 – May 2026

- Led 20-person CAD team in NX Siemens across part modeling, subassembly integration, and technical deliverables for microgravity tool cart handle
- Directed manufacturing and assembly process while ensuring design feasibility and structural performance

Research and Testing Lead

Sep 2024 – May 2025

- Presented "FANG" project at AIAA Region III conference; secured 3rd place out of 60 teams
- Led technical writing and experimental validation of tool performance under simulated conditions

Spaceshot

Structural Design & Analysis Engineer

Aug 2024 – Present

- Performed FEA-driven validation of fin redesign following structural failure, contributing to record 75,124 ft launch
- Identified simulation discrepancies due to launch rail dynamics; developed SRAD rail concept projected to improve altitude prediction accuracy by 17%

Technical Skills

Software: Python, Excel, Ansys Workbench, Nastran, NX Siemens, Java, C++, MATLAB, Ansys Thermal Desktop

Engineering: Trajectory simulation, FEA, CFD, structural analysis, material research, technical writing

Publications

- Design and Manufacturing of FANG (Fabric Anchoring Gadget) for Fabric Repair on the ISS
DOI: 10.2514/6.2025-100003